Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following statement about anhydrous aluminum chloride is correct?

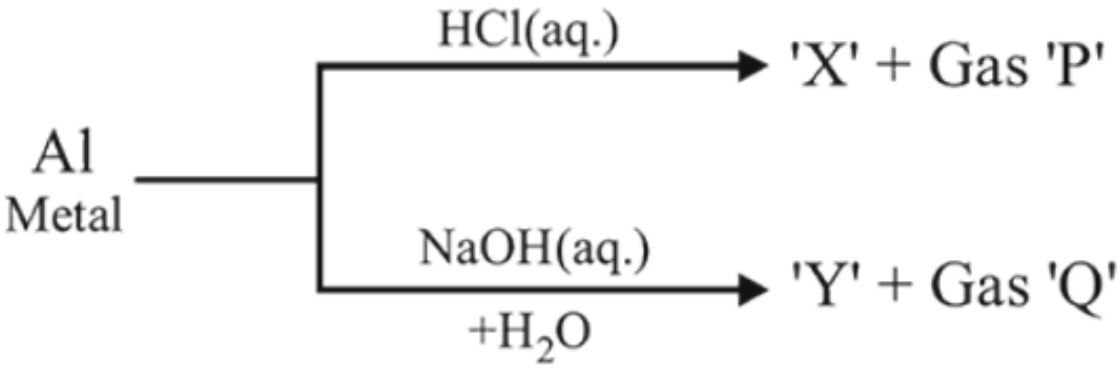
(A) It fumes in moist air.

(B) It exists as dimer both in the vapor state below $350^{\circ}C$ and in non-polar solvents

(C) It is prepared by heating Al_2O_3 in a stream of Sulphur chloride (S_2Cl_2) vapor and chlorine.

(D) All of these

Q.2



The incorrect statement regarding above reactions is:

(A) Al shows amphoteric character

(B) Gas 'P' and 'Q' are different

(C) Both X and Y are water soluble

(D) Gas Q is inflammable

Q.3

In the following sets of reactants which two sets best exhibit the amphoteric characters of $Al_2O_3 \cdot xH_2O$?

Set 1: $Al_2O_3 \cdot xH_2O(s)$ and OH^- (aq)

Set 2: $Al_2O_3 \cdot xH_2O(s)$ and $H_2O(l)$

Set 3: $Al_2O_3 \cdot xH_2O(s)$ and $H^+(aq)$

Set 4: $Al_2O_3 \cdot xH_2O(s)$ and $NH_3(aq)$

(A) 1 and 2

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(B) 1 and 3

(C) 2 and 4

(D) 3 and 4

Q.4

On adding ammonium hydroxide solution to $Al_2(SO_4)_3(aq)$:

(A) A precipitate is formed which does not dissolve in excess of ammonium hydroxide

(B) A precipitate is formed which dissolves in excess of ammonia solution

(C) No precipitate is formed

(D) None of these

Q.5

$AlCl_3$ achieves stability by forming a dimer. In trivalent state the compound is hydrolysed in water. $AlCl_3$ in acidified aqueous solution forms

(A) $Al(OH)_3 + HCl$

(B) $\left[Al(H_2O_6\right]^{3+} + 3Cl^-$

(C) $AlCl_3.2H_2O$

(D) $Al_2O_3 + HCl$

Q.6

The dissolution of $AlOH_3$ by a solution of NaOH results in the formation of:

(A) $[Al(H_2O)_4(OH)_2]^+$

(B) $[Al(H_2O)_3(OH)_3]$

(C) $[Al\,(H_2O)_2(OH)_4]^-$

(D) $[Al(H_2O)_6(OH)_3]$

Q.7

Which statement is not true about potash alum?

(A) On heating it melts and loses its water of crystallization.

(B) It's aqueous solution is basic in nature.

(C) It is used in dyeing industries.

(D) It's empirical formula is $KAl\,(SO_4)_2 \cdot 12H_2O$.

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Q.8

Anhydrous aluminium chloride $\left(Al_2Cl_6\right)$ is covalent compound and soluble in water giving:

(A) Al^{3+} and Cl^- ions

(B) $\left[Al\left(H_2O\right)_6\right]^{3+}$ and Cl^- ions

(C) $\left[AlCl_2^-\left(H_2O\right)_4\right]^+$ and $\left[AlCl_4\left(H_2O\right)_2\right]^-$ ions

(D) none of the above

Q.9

Anhydrous $AlCl_3$ cannot be obtained from which of the following reactions?

(A) Heating $AlCl_3 \cdot 6H_2O$

(B) By passing dry HCl over hot aluminium powder

(C) By passing dry Cl_2 over hot aluminium powder

(D) By passing dry Cl_2 over a hot mixture of alumina and coke

Q.10

In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false ?

(A) CO and CO_2 are produced in this process

(B) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity

(C) Al^{3+} is reduced at the cathode to form Al

(D) Na_3AlF_6 helps in increasing the melting point of the mixture

Q.11

Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives

(A) $\left[Al(OH)_6\right]^{3-} + 3HCl$

(B) $\left[Al\left(H_2O\right)_6\right]^{3+} + 3Cl^-$

(C) $Al^{3+} + 3Cl^-$

(D) $Al_2O_3 + 6HCl$

Q.12

When Al is added to $NaOH$ solution

(A) No action takes place

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(B) $NaAlO_2$ is formed and H_2 is evolved

(C) $Al(OH)_3$ is formed and H_2 is evolved

(D) Na_2AlO_2 is formed and H_2 is evolved

Q.13

The melting pt. of group 13 follows the order

(A) $B > Al > Ga > In > Tl$

(B) $B > Al < Ga > In > Tl$

(C) $B > Al > Tl > In > Ga$

(D) $B > Al < Ga < In < Tl$

Q.14

AlI_3, when reacts with CCl_4, gives

(A) $AlCl_3$

(B) CI_4

(C) Al_4C_3

(D) both (a) and (b)

Q.15

Which of the following properties of aluminium makes it useful for food packaging ?

(A) Good electrical conductivity

(B) Good thermal conductivity

(C) Low density

(D) Non toxicity

Answers & Solutions

Q.1 Answer:

All of these

Solution:

(i) $AlCl_3 + 3H_2O \rightarrow AlOH_3 + 3HCl \uparrow$ fumes

(ii) It exists as dimer in non-polar solvents like benzene and in vapour state below 350^oC **.**

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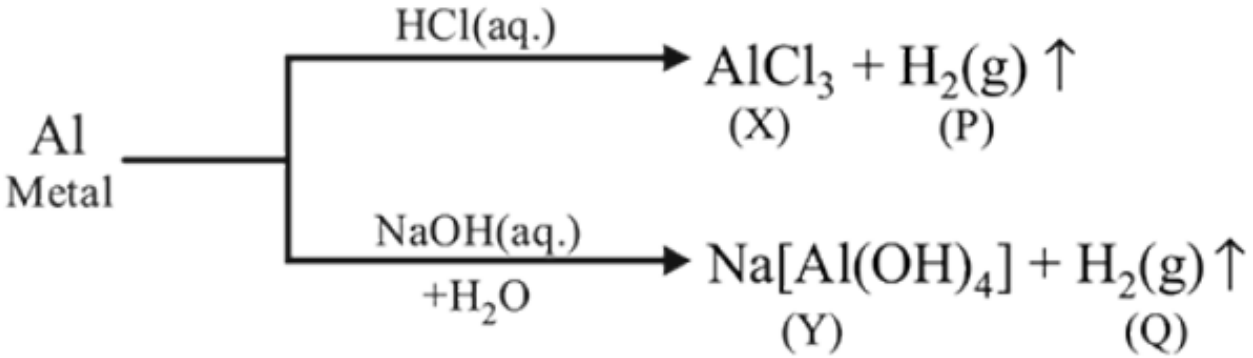
(iii) $4Al_2O_3 + 3S_2Cl_2 + 9Cl_2 \rightarrow 8AlCl_3 + 6SO_2 \uparrow$

A , B , C **are correct** D

Q.2 Answer:

Gas 'P' and 'Q' are different

Solution:



Q.3 Answer:

1 and 3

Solution:

Aluminium oxide is amphoteric oxide because it shows the properties of the both acidic and basic oxides. It reacts with both acids and bases to form salt and water.

$Al_2O_3 \cdot xH_2O + 2NaOH \longrightarrow NaAlO_2 + H_2O$

$Al_2O_3 \cdot xH_2O + HCl \longrightarrow AlCl_3 + H_2O$

Q.4 Answer:

A precipitate is formed which does not dissolve in excess of ammonium hydroxide

Solution:

$$Al_2(SO_4)_3 + 6NH_4OH \longrightarrow 2Al(OH)_3 + 3(NH_4)_2SO_4$$

$$Al(OH)_3 + NaOH \longrightarrow \underset{\text{Soluble complex}}{Na^+[Al(OH)_4]^-}$$

It is insoluble in NH_4OH.

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Q.5 Answer:

$[\ AlH_2O_6^{3+} + 3Cl^-$

Solution:

$AlCl_3$ **in acidified aqueous solution forms** $AlH_2O_6^{3+}$ **ion.**

$$AlCl_3 + 9H_2O \rightarrow AlH_2O_6OH_3 + 3H^+ + 3Cl^-$$

Q.6 Answer:

$Al\ H_2O_2OH_4^-$

Solution:

(1) $[Al(H_2O)_4(OH)_2]^+$

(2) $Al(OH)_3$ **dissolves in** $NaOH$ **solution to give** $Al(OH)^{-4}$ **ion which is supposed to have the octahedral complex species** $[Al(OH)_4(H_2O)_2]^-$ **in aqueous solution.**

(3) $Al(OH)_3 + NaOH(aq) \longrightarrow [Al(OH)_4(H_2O)_2]^-(aq) + NaOH^+(aq)$

Q.7 Answer:

It's aqueous solution is basic in nature.

Solution:

Alum form acidic solution due to hydrolysis of Al^{3+}

Q.8 Answer:

$[AlCl_2^-(H_2O)_4]^+$ and $[AlCl_4(H_2O)_2]^-$ ions

Solution:

$$AlCl_3.6H_2O \xrightarrow{\text{dissociation}} [AlCl_2(H_2O)_4]^+ + [AlCl_4(H_2O)_2]^-$$

Q.9 Answer:

Heating $AlCl_3 \cdot 6H_2O$

Solution:

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$$2Al + 6HCl \xrightarrow{\Delta,\ air} 2AlCl_3 + 3H_2$$

$$2Al + 3Cl_2 \longrightarrow 2AlCl_3$$

$$Al_2O_3 + 3C + 3Cl_2 \xrightarrow{1000\,^\circ C} \underset{\text{vapours}}{2AlCl_3} + 3CO$$

$$\downarrow \text{Cooling}$$

$$\text{Anhydrous } AlCl_3$$

$$AlCl_3.6H_2O \xrightarrow{\Delta} Al(OH)_3 + 3HCl + 3H_2O$$

Thus $AlCl_3$ cannot be obtained by this method.

Q.10 Answer:

Na_3AlF_6 helps in increasing the melting point of the mixture

Solution:

Hall heroult process:

In the Hall-Héroult process the following simplified reactions take place at the carbon electrodes:

Cathode: $Al^{3+} + 3e^{-1} \rightarrow Al$

Anode: $O^{2-} + C \rightarrow CO + 2e^-$

Overall:

$Al_2O_3 + 3C \rightarrow 2Al + 3CO$

And $2\ Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

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In reality, much more CO_2 is formed at the anode than CO :

Cryolite (Na_3AlF_6) lowers the melting point of the mixture and make it more conductive in nature so as to carry out electrolysis of the ore in Hall-Heroult process.

Q.11 Answer:

$[Al(H_2O)_6]^{3+} + 3Cl^-$

Solution:

$$Al_2Cl_6 + 12H_2O \rightleftharpoons 2[Al(H_2O)_6]^{3+} + 6Cl^-$$

Q.12 Answer:

$NaAlO_2$ is formed and H_2 is evolved

Solution:

$$Al + NaOH + H_2O \rightarrow NaAlO_2 + H_2 \uparrow$$

Q.13 Answer:

$B > Al > Tl > In > Ga$

Solution:

Due to structural changes, melting point, increases from Ga to Tl and Ga has the lowest melting point.

Q.14 Answer:

both (a) and (b)

Solution:

AlI_3, on reaction with CCl_4, gives the $AlCl_3$ $4AlI_3 + 3CCl_4 \longrightarrow 4AlCl_3 + 3CI_4$

Q.15 Answer:

Low density

Solution:

Due to the low density of aluminium, it is useful for food packaging.

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